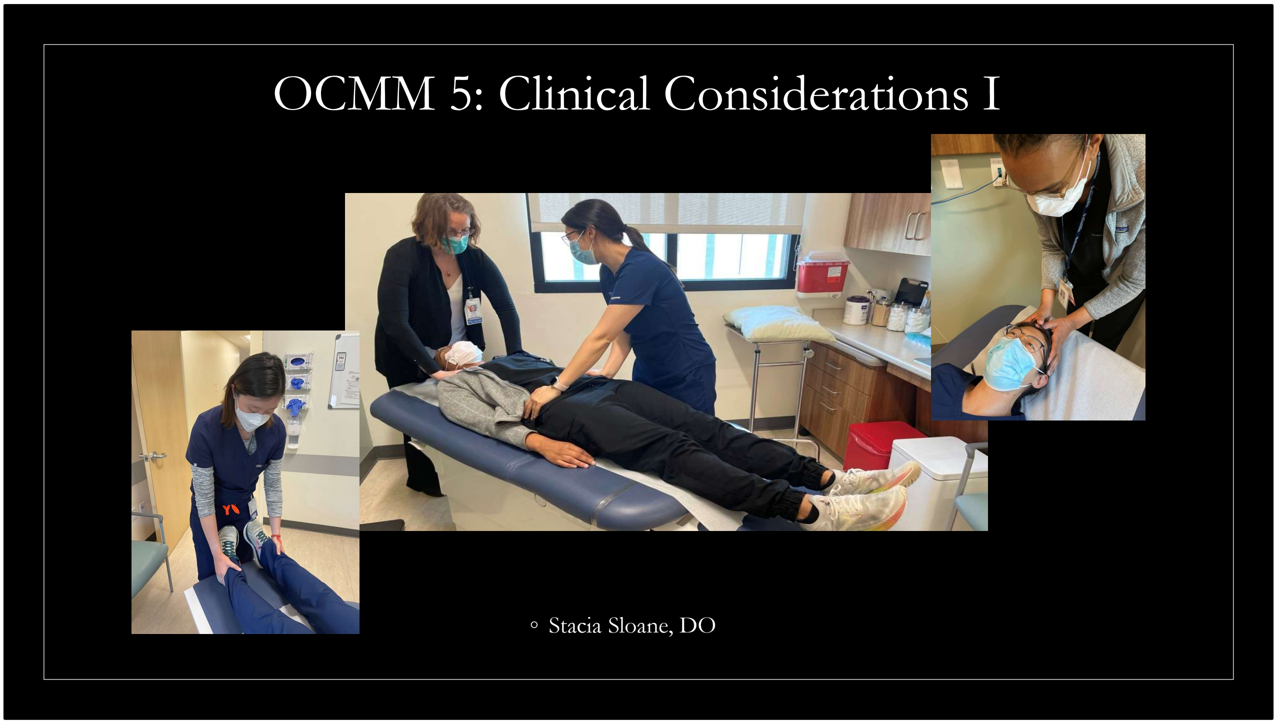



Thank you: Jia-Min Ooi, OMSII and Melody Wang, OMSII as well as Samiyyah Tillman, PGY-2 for your help in clinic at Lifelong and in getting these images.

# Objectives

1. Identify the emphasized cranial, muscular, ligamentous, & neural anatomy related to the pathophysiology of the following clinical cases:
   1. **Headache**
   2. TMJD
   3. Bell's Palsy
   4. Vertigo/Dizziness
2. Define the role of different treatments, including OCMM and other OMT techniques, as it relates to the pathology discussed and emphasized in the self-study
3. Compare and contrast normal motion and pathological motion of the individual cranial structures potentially associated with conditions discussed during the self-study.
4. Define relevant research as it relates to the conditions emphasized in the self-study.

# Case 1

- 32-year-old woman presents to office reporting headaches which she has been getting since she was 15-years-old. The headaches occur 2-3 times per week, however during her menstrual cycle or particularly stressful times the headaches are daily. The pain is described as throbbing, often worse on the left side, and is worse with bright lights, noises and eye activity. The episodes can last up to 8 hours and are often associated with nausea and dizziness, however she denies vomiting, visual changes, numbness/tingling or weakness. She has been on several different medications, both as rescue meds and as daily meds which she has discontinued because of the side-effects. She has had a brain MRI in the past which was normal. She has seen multiple specialists, including a neurologist.
- More recently the headaches have increased slightly in frequency and intensity and our now affecting her daily activities- she has stopped jogging and feels she is less effective at work.
- Meds: None currently
- Allergies: NKDA
- PMHx/PSHx: as above, no surgeries
- Trauma: Car accident at 17 y/o- no LOC
- Family Hx: Mother and Father both alive and well. Mother gets headaches. Father has HTN
- Social Hx: Works as a librarian, 1 cup of coffee in the morning, no tobacco use, no marijuana use, no illicit drugs, no ETOH

“How are the HA affecting your daily life” --- gives major insight into severity. Functionality is key!

# Case 1

- VSS
- PE:
  - HEENT: PERRLA, EOMI, oropharynx clear, TMI, no bulging/fluid, no sinus tenderness
  - Neuro: AAOx3, CNII-XII intact, DTR 2+, MS 5/5 UE and LE, sensation to light touch and pin prick intact, neg Romberg, normal gait including heel-to-toe
- OSE:
  - Cranium: TP on supraorbital notch, sphenoid, squamosal suture on the left and posterior auricular on the right. Right OM suture was compressed, left temporal bone was internally rotated and left torsion pattern of the SBS.
  - Cervical: Flattened Lordosis, OAFSrRl, AARl, C3ESRr, paraspinal muscles ropy
  - Thoracic: T2 FSlRl, paraspinal muscles ropy
  - Ribs: right first rib elevated
  - Lumbar: Increased Lordosis, L5FSrRr
  - Sacrum: LonL FST
  - Pelvis: L Inflare

# Differential Diagnosis:
## Not Comprehensive

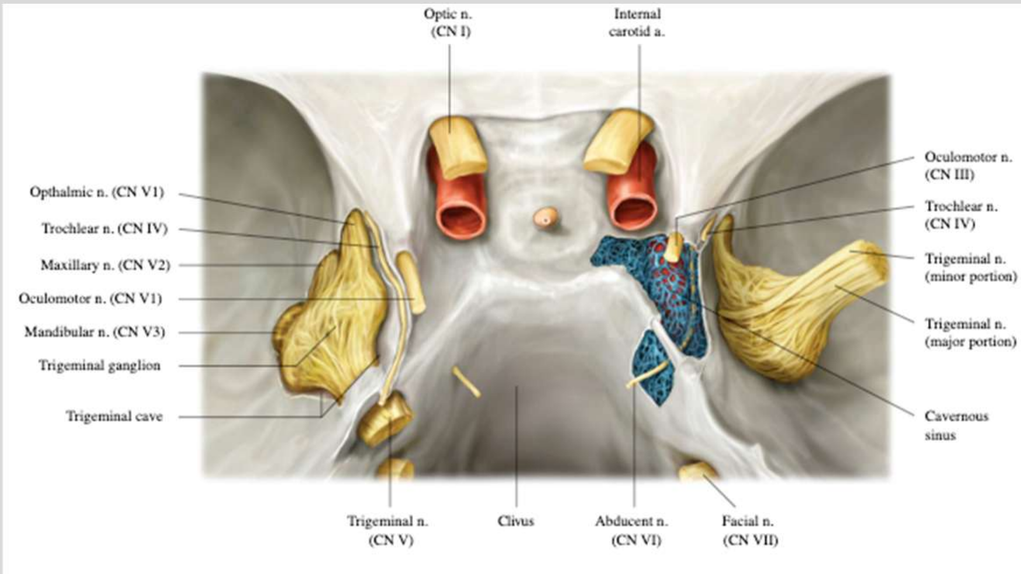
- Migraine
- Cervicogenic
- Tension Type Headache (TTH)
- Sinusitis
- Cluster HA
- AV malformation
- Tumor
- Aneurysm
- Chiari Malformation
- Meningitis
- Pseudotumor Cerebri
- Temporal Arteritis
- MS

# Osteopathic Diagnoses

- Somatic Dysfunction of the Cranium
- Somatic Dysfunction of the Cervical Region
- Somatic Dysfunction of the Thoracic Region
- Somatic Dysfunction of the Ribs
- Somatic Dysfunction of the Lumbar Region
- Somatic Dysfunction of the Sacrum and Pelvis

| **Standing/ Static Exam**: | | **Gait**: | | | |
|---|---|---|---|---|---|
| Musculoskeletal **(TART)** FINDINGS | | DNE* | NSF* | Region Evaluated | Osteopathic Diagnosis |
| | | o | o | Head **(Cr/OA)** | |
| | | o | o | Cervical **(AA/C)** | |
| | | o | o | Thoracic **(T)** | |
| | | o | o | Ribs **(R)** | |
| | | o | o | Lumbar **(L)** | |
| | | o | o | Innominates/ Pelvis **(P)** | |
| | | o | o | Sacrum **(S)** | |
| N/A | | o | o | Lower Extremity **(LE)** | N/A |
| N/A | | o | o | Upper Extremity **(UE)** | N/A |
| | | o | o | Abdomen/ Diaphragm-Other **(ABD)** | |

*DNE= Did Not Examine /*NSF= No Significant Findings  **You must have findings & diagnosis, or one of the circles checked for each region.**

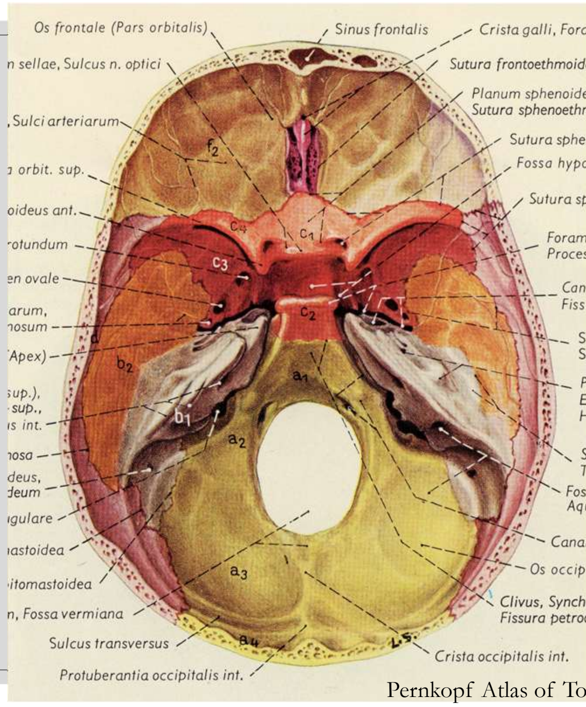



Gilroy, Atlas of Anatomy, 4th ed., Fig. 42.8, Illustrator: M. Voll/K. Wesker,

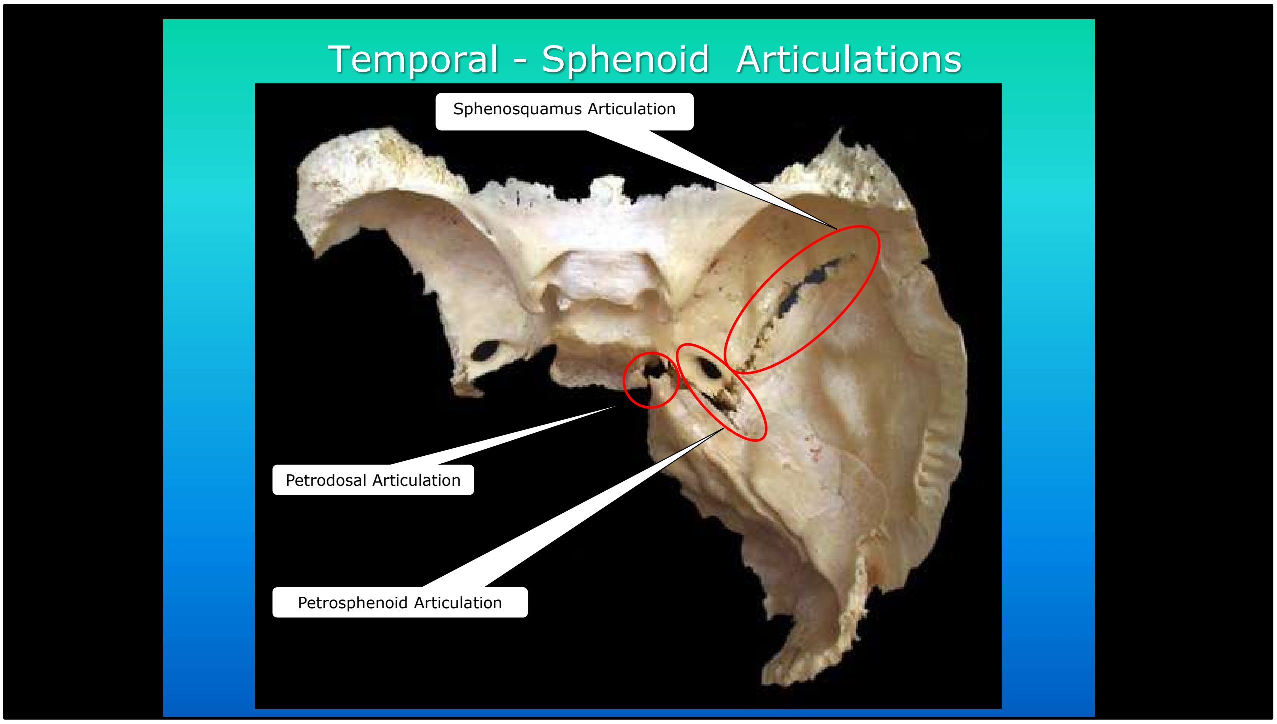

CRANIAL BASE– SUPERIOR VIEW

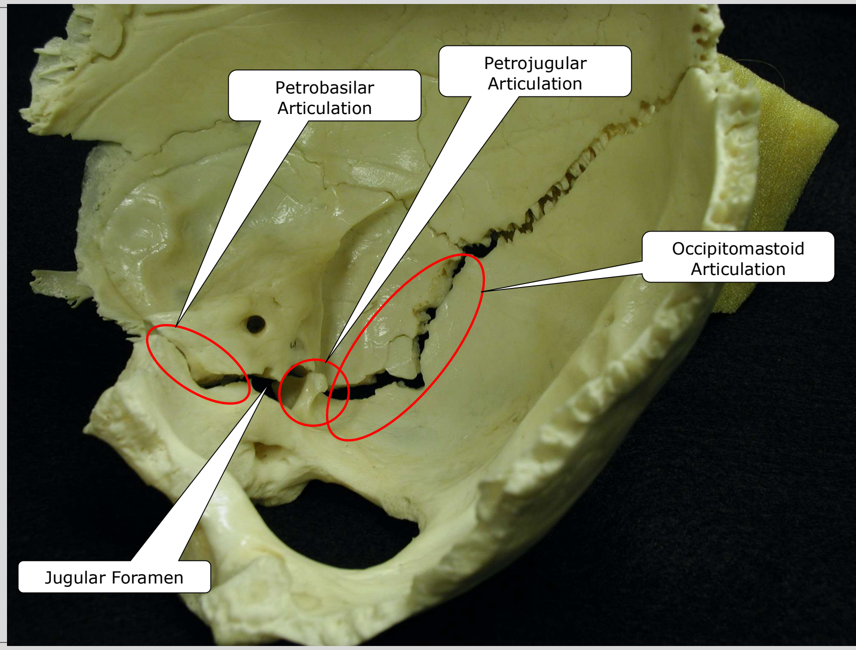
Trigeminal ganglion seems to be the pain generator when it comes to HA– Appreciate Trigeminal ganglion- and its relationship to Meckel's cave- the dural overlay – particularly susceptible to dural strains.

The right ganglion is pulled laterally from its normal position (thereby exposing the trigeminal cave= Meckel's cave) to show the cavernous sinus and the internal carotid artery which passes through the sinus.

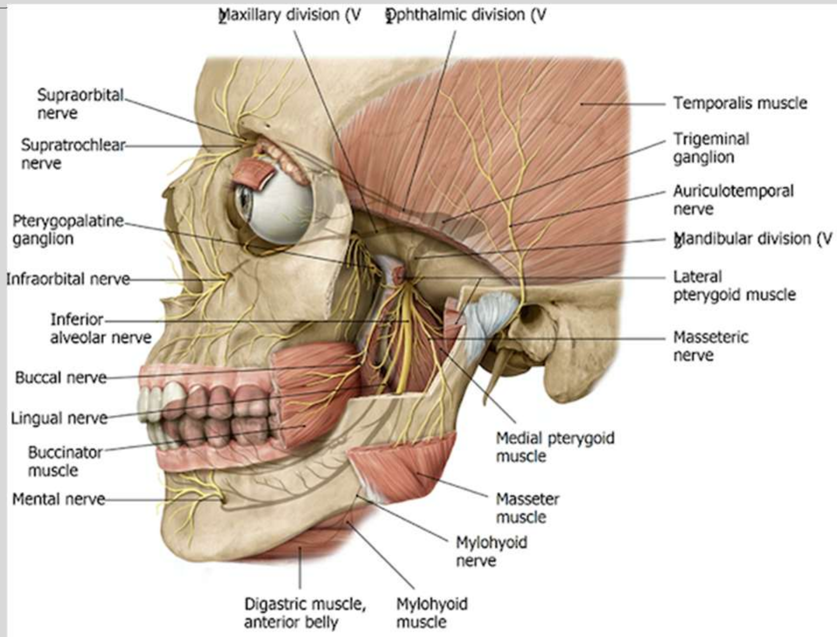


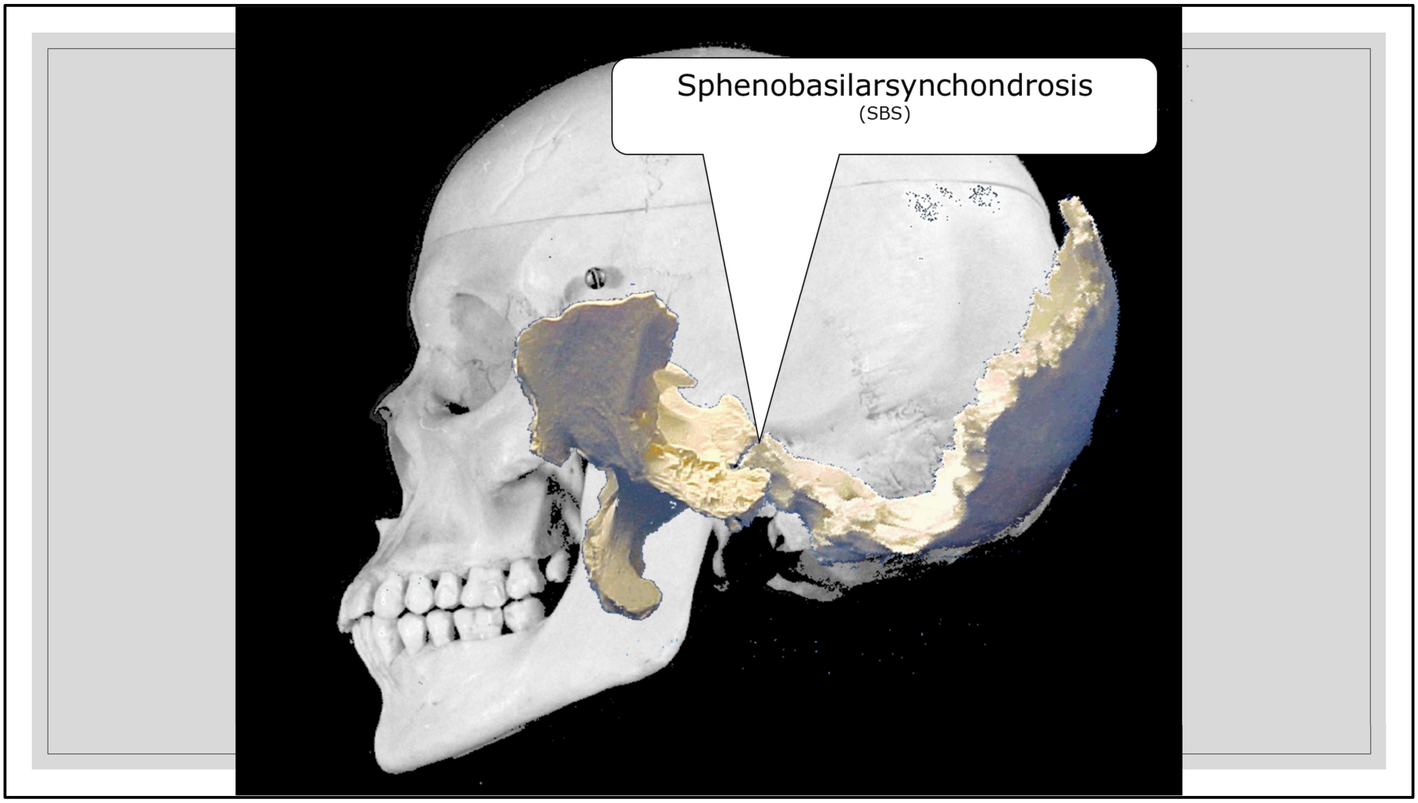

Pernkopf Atlas of Topographic and Applied Human Anatomy


Temporal - Sphenoid Articulations
Sphenosquamus Articulation
Petrodosal Articulation
Petrosphenoid Articulation



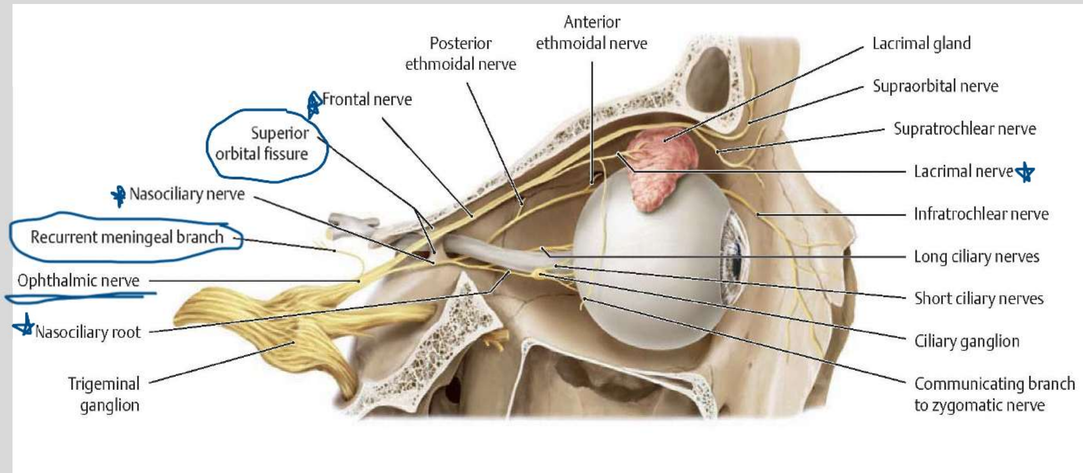
Petrobasilar
Articulation
Petrojugular
Articulation
Occipitomastoid
Articulation
Jugular Foramen


Maxillary division (V
Ophthalmic division (V
Supraorbital nerve
Supratrochlear nerve
Pterygopalatine ganglion
Infraorbital nerve
Inferior alveolar nerve
Buccal nerve
Lingual nerve
Buccinator muscle
Mental nerve
Digastric muscle, anterior belly
Mylohyoid muscle
Mylohyoid nerve
Masseter muscle
Medial pterygoid muscle
Masseteric nerve
Lateral pterygoid muscle
Mandibular division (V
Auriculotemporal nerve
Trigeminal ganglion
Temporalis muscle




Flexion

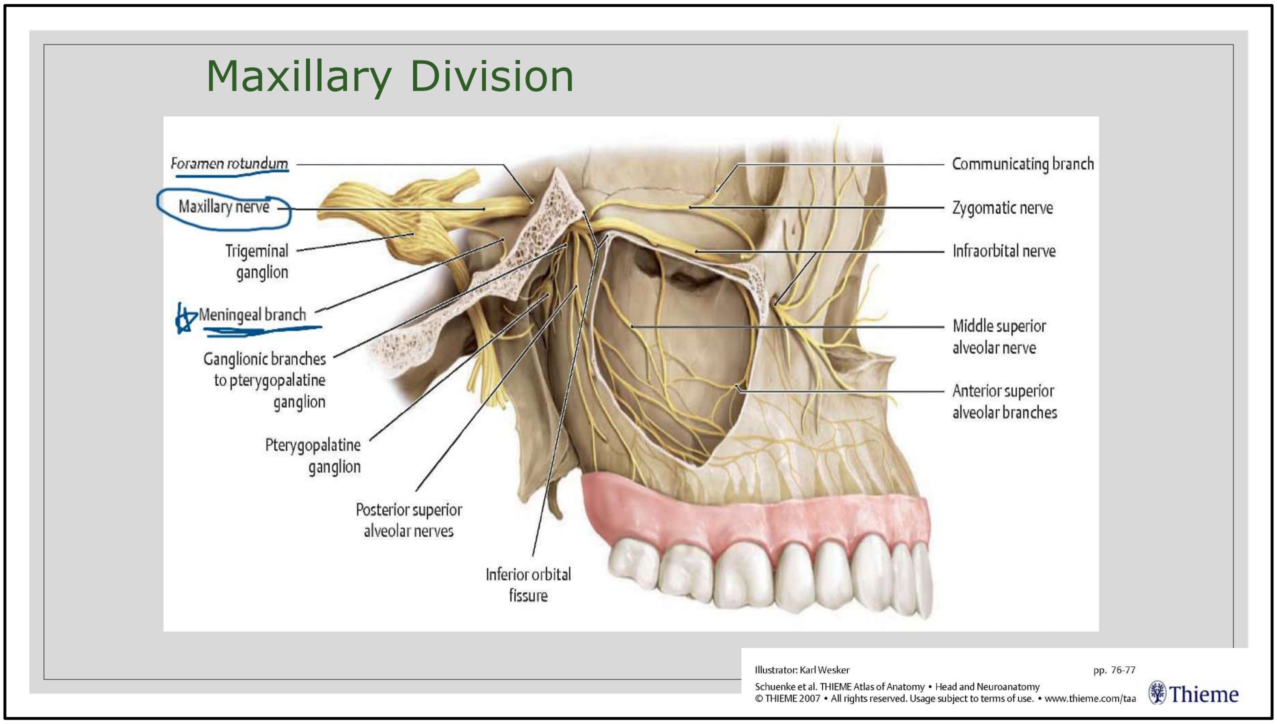
## Ophthalmic Division






***Trigeminal nerve sends out divisions that innervate the dura.***

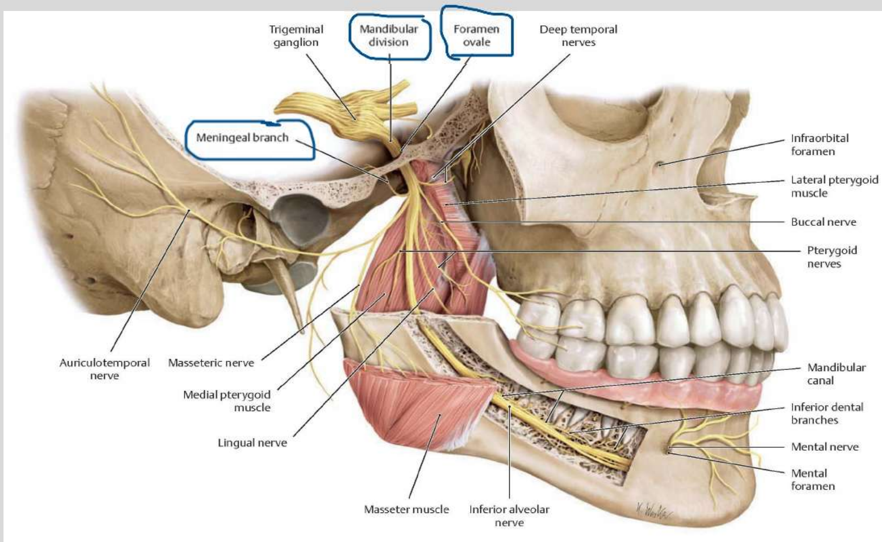
Ophthalmic Division ($V_1$) –Proceeds from the trigeminal ganglion and promptly branches off via the recurrent meningeal nerve to travel through the lateral wall of the cavernous sinus to deliver sensory from the **anterior cranial fossa dura mater**. The portion that carries ontravels through through the **superior orbital fissure** to divide into the nasociliary, frontal, & lacrimal nerves respectively




*Maxillary Division ($V_2$) –*

Proceeds from the trigeminal ganglion over the **sphenosquamous articulation** and then sends out a meningeal branch delivering sensory from the **middle cranial fossa dura.** Then it travels on through **foramen rotundum** and directly lateral to the orbital process of the palatine bone. Next it divides to form the zygomatic, infraorbital, and alveolar nerves with communications to the **pterygopalatine ganglion** in the pterygopalatine (infratemporal) fossa

## Mandibular Division



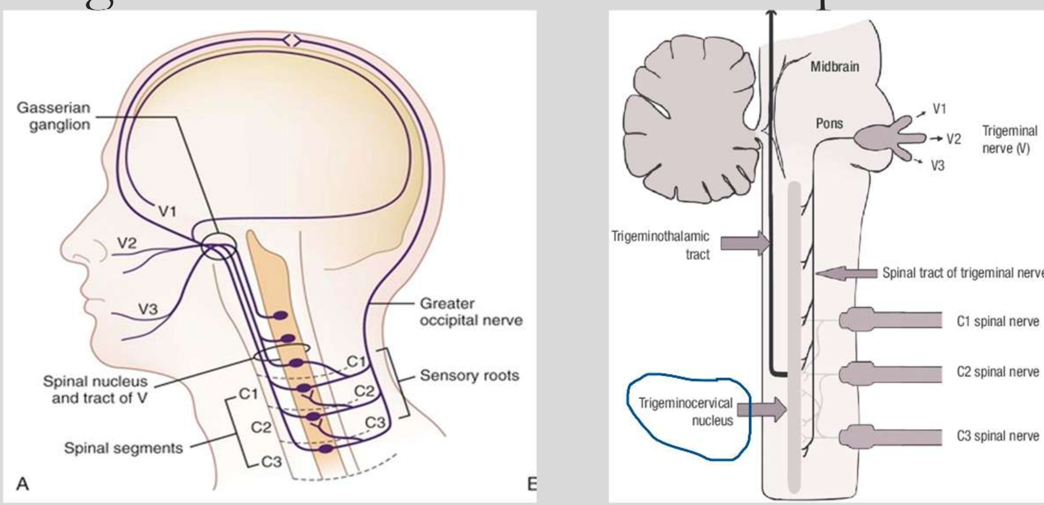

Illustrator: Karl Wesker pp. 76-77
Schuenke et al. THIEME Atlas of Anatomy • Head and Neuroanatomy


Mandibular Division ($V_3$) –

Proceeds over the **sphenosquamous articulation** then through **foramen ovale**. **After which it too gives off a sensory meningeal branch** that renters through foramen spinosum to innervate the **middle cranial fossa dura.**

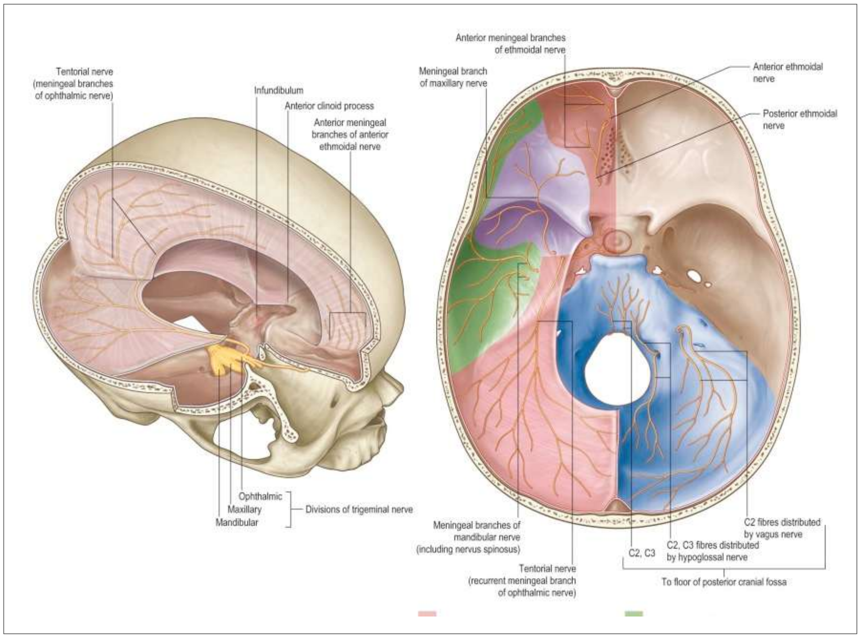
$V_3$ then proceeds into the infratemporal fossa splaying forth its **sensory and motor** branches to the dentition, mucosa, auditory areas and skin as well as motor for muscles of mastication

# Trigeminal Nerve and Cervical Spine




https://www.researchgate.net/figure/Mechanism-of-cervicogenic-headache-pain-referred-pain-from-cervical-structures-due-to_fig2_299434211

The trigeminal nerve in the brain stem extends and becomes continuous with the grey matter of the spinal cord, becoming the trigeminocervical nucleus. The nucleus receives afferent fibers from not only the trigeminal nerve, but from fibers of the VII, IX, and X cranial nerves. This ramification extends to the 3rd or fourth cervical segment of the spinal cord. The significance of this nucleus is that all nociceptive afferent fibers to the head and neck exist in the specific area and have overlapping patterns of innervation with each other. **This creates the basis for referred pain in the head and neck.** For example, pain in the head (CN V) can be referred back to a cervical or other cranial nerve segment resulting in pain in a different receptive field, or both fields.




# Dural Innervation

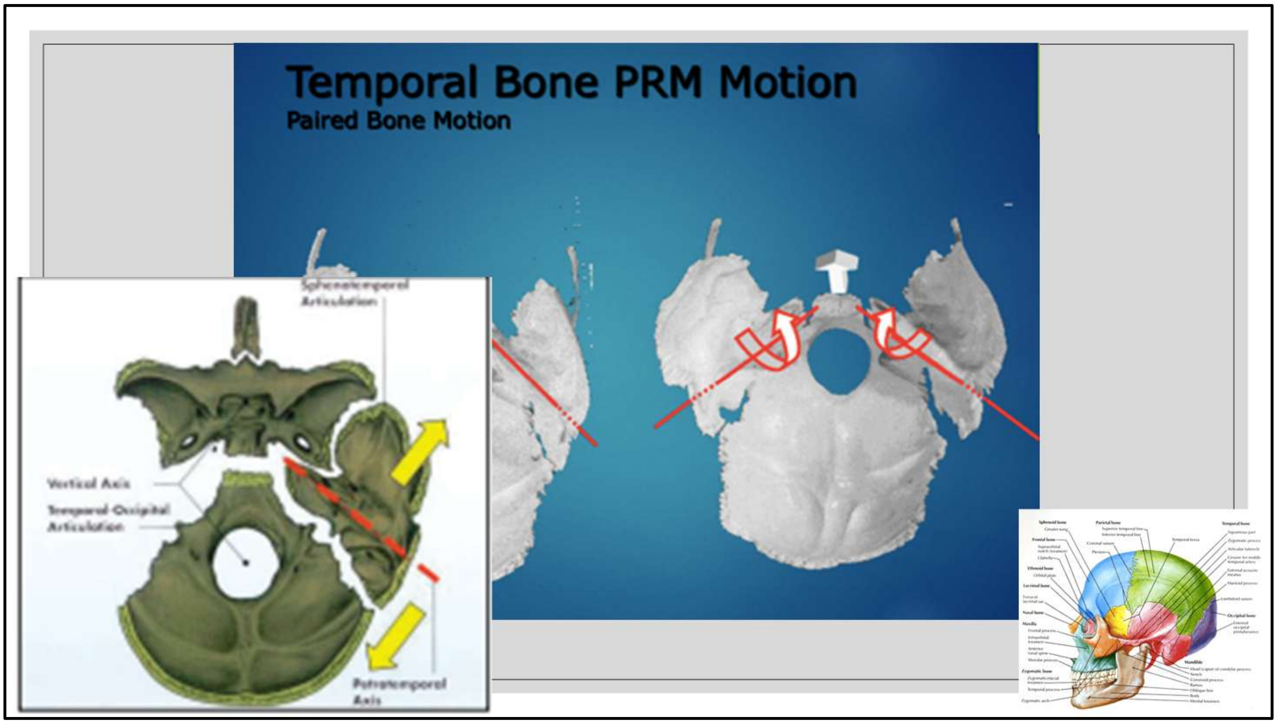
Posterior fossa is innervated by fibers of the upper cervical spine with the vagus nerve
Middle and anterior cranial fossa- innervated by branches of the trigeminal nerve (V1,2,3)
Falx cerebri and the tentorium cerebelli- innervated by the trigeminal nerve branches, especially recurrent meningeal which branches off of V1.

# Headaches

- Relevant Anatomy:
  - C1-C3 and the Trigeminal Spinal Tract (V, IX and X)
  - T1-4 Sympathetic Innervation to Head and Neck
  - Suboccipital Muscles
  - Meckel's cave is susceptible to dural strains
  - Sphenosquamous articulation ($V_2$ & $V_3$)
  - Temporal and Sphenoid Bone Somatic Dysfunction (petrosphenoidal articulation- Meckel's Cave)
  - Dura including falx cerebri and tentorium cerebelli
  - Superior orbital fissure (sinus HA)
  - Pelvis and Lower Extremities
  - Spine (Scoliosis)

Upper cervical vertebra- relationship to trigeminal spinal tract, and their innervation of posterior cranial fossa
T1-4 sympathetic innervation to head and neck– increased sympathetic tone intensifies muscle irritability
Suboccipital muscles- close relationship to the head and neck, also often related to upper thoracic and rib dysfunction/postural decompensation. Sphenosquamous articulation- where the temporal and the sphenoid meet– very significant clinically. Bevel change here. Sutherland discusses this as a critical to the treatment of migraines.

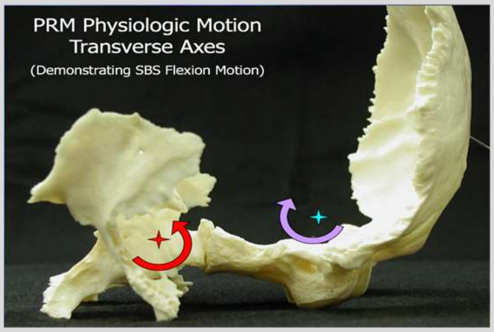
Increased sympathetic tone intensifications muscle irritability.

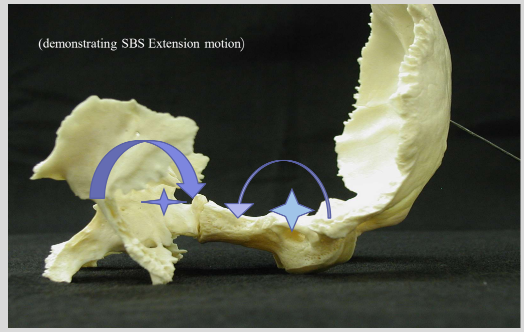



Superior view, looking down at the temporal bones (parietals and frontals removed). Appreciate the axis of rotation traversing through the petrous ridge of the bone– anteriorly converging.
On the right you can see that the basiocciput is rising– as we would see with FLEXION. The temporal bones are being driven by the occiput into EXTERNAL ROTATION– Appreciate how this would cause a relative widening of the anterior temporal area- creating an increased transverse diameter.

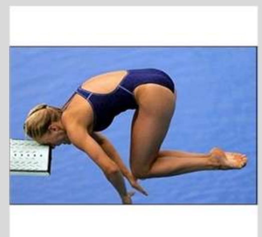
Red line shows us the axis of rotation of the temporal bone. The arrows show us the widening of the anterior-superior portion of the bone in external rotation, and the narrowing of the mastoid portion below the axis. Keep this in mind as you begin to wrap your heads around the shape of the head one would expect in flexion/extension of the midline bones.

figure from “Cranial Strains and Malocclusion: A Rationale for a New Diagnostic and Treatment Approach” By Gavin A. James, MDS, FDS and Dennis Strokon, DDSfrom Orthopedic Gnathology, Hockel, J., Ed. 1983.




# Flexion and Extension



Reminder of normal motion
Review intro to diagnosis and treatment self study--- review all of the potential strain patterns

# SBS STRAIN PATTERNS

**Physiologic-**
***Adaptive***

- **Flexion/Extension** (strain if restricted into either direction)
- **Torsion**
- **Sidebending Rotation**

NON-Physiologic-
*maladaptive*

- Compression
- Lateral Strains
- Vertical Strains

**Physiologic strains "are common and are considered physiologic if their presence does not interfere with the flexion -extension motion of the mechanism" FOM**

Strains, especially non-physiologic, can impair the PRM (primary respiratory mechanism) in doing its work- this may affect overall health of the nervous system

Physiologic strains- like forward sacral torsions- they can be the problem- or just a finding and not really causing any issue with the motion of the sacrum.

# Research- Cost of Migraine Headaches

- Electronic medical records from 631 patients, representing 1427 migraine-related office visits, were analyzed. **Average cost per patient visit was approximately 50% less at the osteopathic clinic than at the allopathic clinic ($195.63 vs $363.84, respectively; *P*.001).** This observed difference was entirely attributable to the difference in the average number of medications prescribed per visit at the two clinics, with 0.696 prescriptions at the osteopathic clinic and 1.285 prescriptions at the allopathic clinic (*P*.001). This difference in prescription number resulted in a lower average medication cost per visit at the osteopathic clinic than at the allopathic clinic **($106.94 vs $284.93, respectively; *P*.001**). Patients at the osteopathic clinic were 5 years younger on average than at the allopathic clinic (*P*.001). No statistically significant difference was observed between the two practices in patients' ratings of pain severity.
- Schaber, E. Crow, T. *Impact of Osteopathic Manipulative Treatment on Cost of Care for Patients With Migraine Headache: A Retrospective Review of Patient Records. J Am Osteopath Assoc.* 2009;109:403-407.

Migraine HA are a common cause of functional disability in the US. The annual cost of lost productivity in the US is estimated to range from $1.2 to $17.2 billion. Missed days of work and cost of medications are both a huge cost to our society.

# Research- Manual Therapy and TTH

- Eighty-four patients diagnosed with tension- type headache (TTH) were randomly assigned to 1 of 4 groups: (1) suboccipital soft tissue inhibition (SI), (2) occiput-atlas-axis (OAA), (3) combined SI and OAA, or (4) placebo. For the SI therapy, a physiotherapist placed his or her hands in contact with the patient's suboccipital muscles in the area of the posterior arch of the atlas and applied steady pressure to release muscle spasm. For the OAA therapy, a physiotherapist administered global manipulation bilaterally, first performing cephalic decompression and then small circumductions on a vertical axis through the odontoid process of the axis. Patients in the placebo group received no treatment and rested in the supine position for 10 minutes. Patients in all groups participated in 4 weekly intervention sessions. Before each session, all patients underwent the vertebral artery challenge test.
- The outcome measures were based on results of the Headache Impact Test-6; the Headache Disability Inventory; headache pain intensity, which was rated daily by patients on a 0- to 10-point visual analog scale; and craniocervical range of motion, which was measured with a standardized physiotherapy device. All outcome measures were collected at base- line, at the conclusion of the 4 weekly intervention sessions, and at an 8-week follow-up visit.
- **At the end of the 4 intervention sessions, statistically significant results were found for the SI, OAA, and combined intervention groups compared with the placebo group on almost all of the outcome measures. Improvements were maintained at the 8-week follow-up evaluation.**
- Espí-López GV, Gómez-Conesa A, Gómez AA, Martínez. JB, Pascual-Vaca ÁO, Blanco CR. Treatment of tension-type headache with articulatory and suboccipital soft tissue therapy: a double-blind, randomized, placebo-controlled clinical trial [published online January 27, 2014]. *J Bodyw Mov Ther*. 2014. doi:10.1016/j.jbmt.2014.01.001.

Done by physiotherapists

# Research- TTH

- **Patients:** Forty-four patients who were affected by frequent episodic TTH and not taking any drugs for prophylactic management of episodic TTH were recruited.
- **Interventions:** Patients were randomly allocated to an experimental or control group. The experimental group received corrective OMT techniques, tailored for each patient; the control group received assessment of the cranial rhythmic impulse (sham therapy). The study included a 1-month baseline period, a 1-month treatment period, and a 3-month follow-up period.
- **Main Outcome Measures:** The primary outcome was the change in patient-reported headache frequency, and secondary outcomes included changes in headache pain intensity (discrete score, 1 [lowest perceived pain] to 5 [worst perceived pain]), over- the-counter medication use, and Headache Disability Inventory score.
- **Results:** Forty patients completed the study (OMT, n=21; control, n=19). **The OMT group had a significant reduction in headache frequency over time that persisted 1 month (approximate reduction, 40%; *P*<.001) and 3 months (approximate reduction, 50%; *P*<.001) after the end of treatment. Moreover, there was an absolute difference between the 2 treatment groups at the end of the study, with a 33% lower frequency of headache in the OMT group (*P*<.001).**
- **Conclusion:** This feasibility study **demonstrated the efficacy of OMT in the management of frequent episodic TTH, compared with sham therapy in a control group.** Osteopathic manipulative therapy may be preferred over other treatment modalities and may benefit patients who have adverse effects to medications or who have difficulty complying with pharmacologic regimens. This protocol may serve as a model for future studies.


- Rolle, G. Tremolizzo, L. Somalvico, F. et al. *Pilot Trial of Osteopathic Manipulative Therapy for Patients With Frequent Episodic Tension-Type Headache. J Am Osteopath Assoc.* 2014;114(9):678-685 doi:10.7556/jaoa.2014.136.


Done in Europe- manual therapy vs. control group.

# Research- Migraines and Mood

- **Context**: The substantial functional impairment associated with migraine has both physical and emotional ramifications. Mood disorders are often comorbid in patients with migraine and are known to adversely affect migraine activity.
- **Objectives**: To explore the effects of osteopathic manipulative therapy (OMT; manipulative care provided by foreign-trained osteopaths) on pain and mood disorders in patients with high-frequency migraine.
- **Methods**: Retrospective review of the medical records of patients with high- frequency migraine who were treated with OMT at the Headache Istituto di Ricovero e Cura a Carattere Scientifico Fondazione Santa Lucia from 2011 to 2015. Clinical assessments were made using the Headache Disability Inventory (HDI), the Headache Impact Test (HIT-6), the Hamilton Depression Rating Scale (HDRS), and the State-Trait Anxiety Inventory (STAI) forms X-1 and X-2.
- **Results**: Medical records of 11 patients (6 women; mean age, 47.5 [7.8] years) with a diagnosis of high-frequency migraine who participated in an OMT program met the inclusion criteria and were included in the study. When the questionnaire scores obtained at the first visit (T0) and after 4 OMT sessions (T1) were compared, significant improvement in scores were observed on STAI X-2 (T0: 43.18 [2.47]; T1: 39.45 [2.52]; P<.05), HIT-6 (T0: 63 [2.20]; T1: 56.27 [2.24]; P<.05), and HDI (T0: 58.72 [6.75]; T1: 45.09 [7.01]; P<.05).
- **Conclusion**: **This preliminary study revealed that patients with high-frequency migraine and comorbid mood disorders showed significant improvement after four 45-minute OMTh sessions. Further investigation into the effects of OMTh on pain and mood disorders in patients with high-frequency migraine is needed**.


- D'Ippolito, M. Tramontano, M. Gabriella Buzzi, M. *Effects of Osteopathic Manipulative Therapy on Pain and Mood Disorders in Patients With High-Frequency Migraine.* J Am Osteopath Assoc. 2017;117(6);365-369 doi:10.7556/jaoa.2017.074


Many patients with migraine have mood disorders.

# Objectives

1. Identify the emphasized cranial, muscular, ligamentous, & neural anatomy related to the pathophysiology of the following clinical cases:
    1. Headache
    2. **TMJD**
    3. Bell's Palsy
    4. Vertigo/Dizziness
2. Define the role of different treatments, including OCMM and other OMT techniques, as it relates to the pathology discussed and emphasized in the self-study
3. Compare and contrast normal motion and pathological motion of the individual cranial structures potentially associated with conditions discussed during the self-study.
4. Define relevant research as it relates to the conditions emphasized in the self-study.

# Case 2

- CC: Jaw Pain
- HPI: 35-year-old male present with right sided jaw pain and clicking. Patient has had intermittent pain for years and has been told he has "TMJ" however 3 weeks ago the symptoms worsened after having dental work done. Patient reports worsening symptoms with chewing and when he wakes up in the morning. He periodically wears a night guard. Patient also reports ear pain and neck pain. Patient has taken ibuprofen with minimal relief. Patient denies any previous trauma.

# Case

- PMHx: History of GERD, no previous hospitalizations or surgeries. Patient did have braces as a child.
  - Meds: Ibuprofen 400mg as needed for jaw pain.
  - Allergies: NKDA
- Family Hx: Non-contributory
- Social Hx: Stock Broker (very stressful), drinks 1-2 beers/day, exercises regularly, SAD diet, 10 pack per year smoking history (current smoker).
- ROS: No fever, chills, or unintended weight loss. No HA, vision changes, numbness/tingling/weakness. No n/v/d/c. No CP or SOB. No urinary or bowel changes. Positive symptoms as per HPI.

Miyawaki, S. et. al. *Association Between Nocturnal Bruxism and Gastroexophageal Reflux*. Sleep Medicine Disorders. http://www.journalsleep.org/Articles/260716.pdf.

SAD- Standard American Diet

# Jaw/Face Pain Differential Diagnosis:
Not Comprehensive

- Temporomandibular Joint Disorder
- Dental Infection
- Mandibular fracture
- Gout, pseudogout, RA
- Tension or migraine headaches
- Otitis Media or Sinusitis
- Temporal arteritis
- Trigeminal neuralgia, postherpetic neuralgia
- Muscle trigger points (SCM, pterygoids, masseter).

# History and physical

- History:
  - Facial Pain
  - Tenderness around joint/muscles.
  - ROM restriction
  - Headaches
  - Neck pain, shoulder pain.
  - Earache, tinnitus, fullness.
  - Jaw trauma (tooth extraction); malocclusion.
  - Bruxism/clenching
  - Click, pop, snapping and/or locking.
  - Throat pain, pain on swallowing or speaking.

- Physical:
  - Visual
    - Asymmetry
    - Muscle Hypertrophy
    - Malocclusion/dental wear
    - Mandible ROM/jaw opening (50 mm vertical/10 mm lateral)
    - Cervical/thoracic spine (scoliosis).
  - Palpatory
    - TMJ and surrounding muscles
    - Tender points/trigger points (pterygoid and masseter refer to TMJ): scalene, SCM, temporalis and suprahyoid.
    - Cranial strains (very common)
    - OSE:
      - Cranial, cervical, thoracic, lumbar, sacrum, pelvis, lower and upper extremity, ribs.

# Work Up

- Lab:
  - CBC (infection), uric acid, ESR, RA, ANA, Ca, Alk Phos.
- Imaging
  - Typically not indicated
  - X-ray (DJD), MRI (meniscal tear-?surgery), CT (bone disease)
- Dental
  - Dental tape and casts (analyze occlusion strain/stress).
  - Orthodontic strain analysis.

# TMJD (Jaw pain)

- TMJ dysfunction is the most frequent source of facial pain after toothache.
  - 10 million people in US.
  - 20-40 year olds; female to male 4:1.
- 4 categories:
  - Myofascial pain dysfunction (MPD) syndrome
    - Psychophysiologic disease
    - Muscles of mastication
    - S.D. elsewhere in the body
  - Internal derangement (ID) injury
    - Abnormal relationship between the articular disc and the mandibular condyle
    - Acute disc displacement and chronic recurrent dislocation.
  - Accessory ligament dysfunction (ALD)
    - Accessory ligaments that help stabilize the joint.
    - Sphenomandibular, stylomandibular and pterygomandiublar.
  - Degenerative joint disease (DJD)
    - Organic degeneration of the articular surfaces within the TMJ.

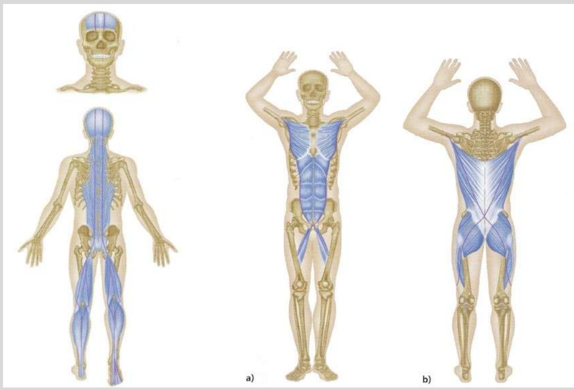
# TMJD (Jaw pain)

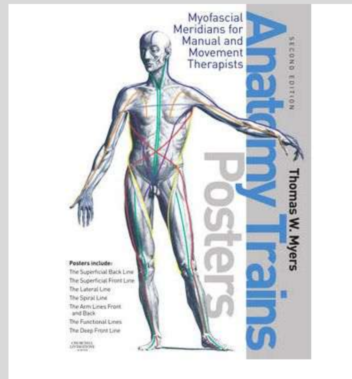
- Etiology and pathophysiology:
  - MPD (myofascial pain dysfunction)
    - Most common cause.
    - Somatic asymmetries leading to malocclusions, jaw clenching, bruxism, psychological symptoms, etc. (muscle hyperactivity/spasm).
    - Whiplash and other thoracic/cervical strains are often overlooked.
    - TMJ patients often show weakness in deep cervical flexor muscles.
    - Dental procedures and anxiety around dental procedures.
  - ID (internal derangement)
    - Biomechanical problem within the joint (muscle spams a result of the problem, not the cause).
    - Anterior disc displacement ("click" and "locked" jaw).
  - ALD (accessory ligament dysfunction)
    - Caused by direct or indirect trauma to mandible (dental, stress).
    - Cranial strains (stress on ligaments).
  - DJD (degenerative joint disease)
    - Trauma to discs, mandibular condyles or connective tissue.
    - OA, RA, ankylosis, infections, neoplasm.

# TMJD (Jaw pain)

- TMJ Joint:
  - One of the most complex joints in body.
  - Hinge joint with gliding motion.
  - Convex articular surface anteriorly and concave posteriorly.
  - Articular disc is biconcave (three parts)
  - Motion: Suprahyoid muscles (opening), lateral pterygoid (gliding), temporalis, masseter and medial pterygoid (closing).
  - Complex muscle contractions with grinding and protraction.

# Kinetic Chain/Anatomy Trains



## Somatic Dysfunctions found in patients with TMD (percentages)

- Cranial dysfunction of some kind – “nearly all”
- OA – 9%
- C3-C4 dysfunction – 50%
- C4-5 – 30%
- T2-3 – 14%
- T3-4 – 14%
- Lumbosacral junction – 29%
- Sacroiliac joints – 32%
- Scoliosis – 14%

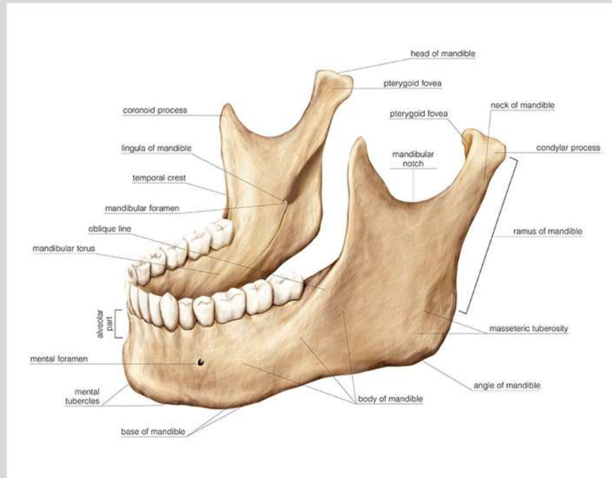
Blood, SD, 1986 (6)

Consider the whole body! This is our unique perspective.

# ANATOMY AND FUNCTION

# The Mandible

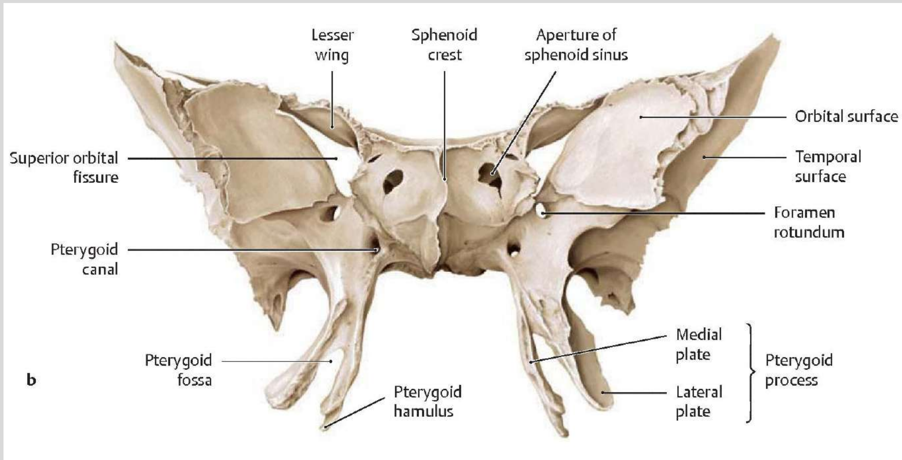
- Condyles
- Coronoid process
- Ramus
- Angle
- Body
- Symphysis Menti (union of two halves)
- Not a part of either neurocranium or viscerocranium- develops from membranous ossification.




https://pixels.com/featured/4-mandible-asklepios-medical-atlas.html

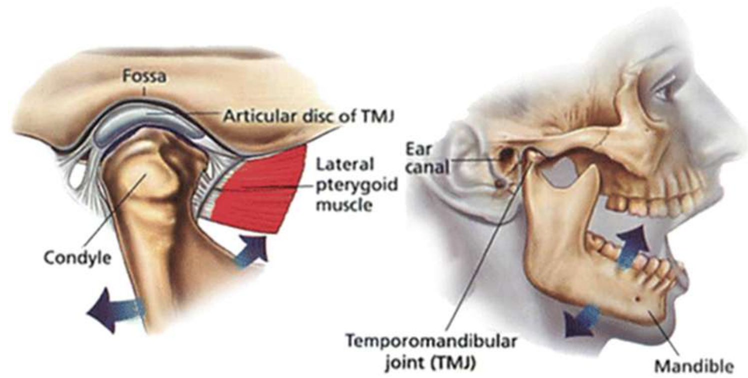
# Sphenoid Bone

Anterior



## Temporomandibular Joint Anatomy

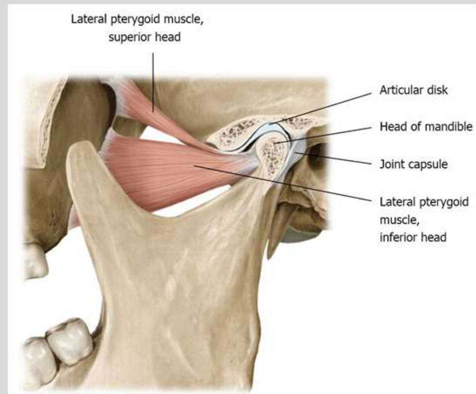
- Compound synovial joint.
- Contains non-vascular fibrous connective tissue.
- Condyle of mandible, fossa of temporal bone, disc in the middle.
- Disc moves forward with condyle of mandible.
- Lateral pterygoid connects into the disc.




https://www.google.com/url?sa=i&source=images&cd=&ved=2ahUKEwjB3PL6xMnjAhUiqlQKHRZwBgcQjRx6BAgBEAU&url=https%3A%2F%2Flink.springer.com%2Fchapter%2F10.1007%2F978-94-017-9804-4_49&psig=AOvVaw2aqu2WlxxmFUKlKuLZa09a&ust=1563919674421495

# Anatomy: TMJ Disc

•Articular disc divides joint into upper and lower compartments.
•Thin in middle; thick on both ends.
•Disc connected to joint capsule anteriorly.
•Posteriorly disc becomes continuous with the posterior attachment tissues.
•Lateral Pterygoid attaches to disc




https://www.google.com/url?sa=i&source=images&cd=&ved=2ahUKEwiC_5D7xcnjAhWpi1QKHcATD3wQjRx6BAgBEAU&url=https%3A%2F%2Ftreningogrehab.no%2Ftrue-cause-solution-temporomandibular-dysfunction-tmd%2F&psig=AOvVaw0_PocU6YgVM7BZtfO51q3b&ust=1563919926568346

# Normal TMJ Function

•Requires intact joint and balanced, coordinated muscular function

**Initial opening (A-B in figure):** rotation of condyle against inferior surface of disk (20-30mm of opening).

**Continued opening (C-D in figure):** translatory motion of disk and condyle together as a unit, within superior synovial joint.

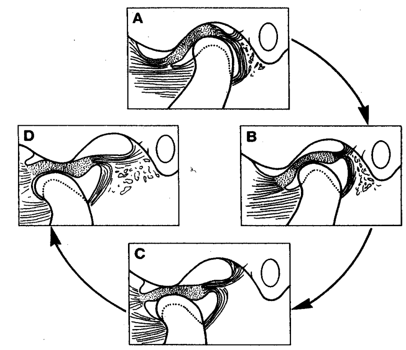
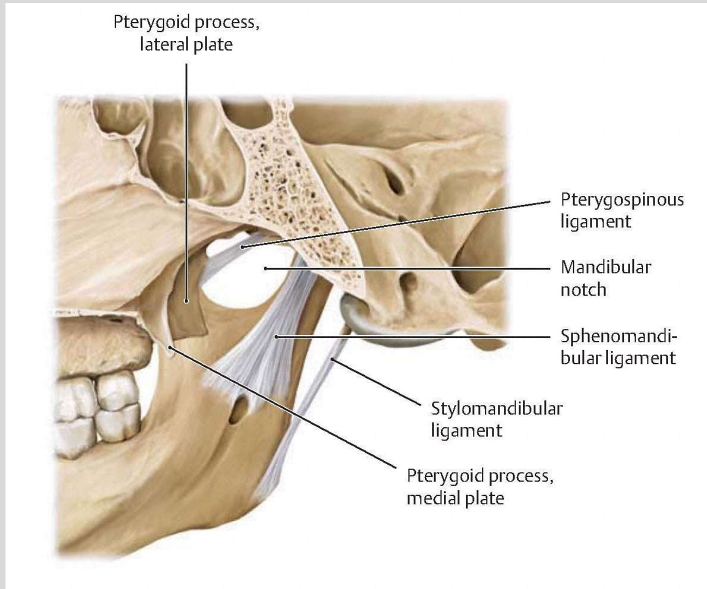



FIGURE 45.26. Opening motion of temporomandibular joint. (After Blaschke DD, from Solberg WK, Clark GT. *Temporomandibular Joint Problems: Biologic Diagnosis and Treatment*. Chicago, IL: Quintessence Publishing; 1980:73, with permission.)

•**Motion Limiting:** Posteriorly, the superior lamina is stretched and the inferior lamina limits end-range of motion (D).

## TMJ Ligaments
(Deep Medial to Lateral View)




Reference 1

Important to consider when treating! BLT in upcoming labs.

# Normal TMJ Function

•Total normal adult opening 50 mm.

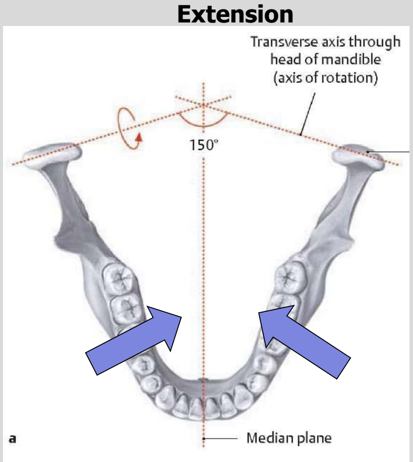
•Side-to-side movement is normal at 10-12 mm.

- <40 mm opening = hypomobile
- >70 mm opening = hypermobile

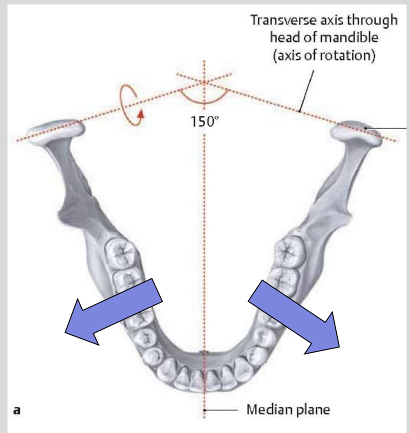
Both hypo and hypermobility can lead to dysfunction.

# Mandible Gross Versus Primary Respiratory Motion

**Extension**



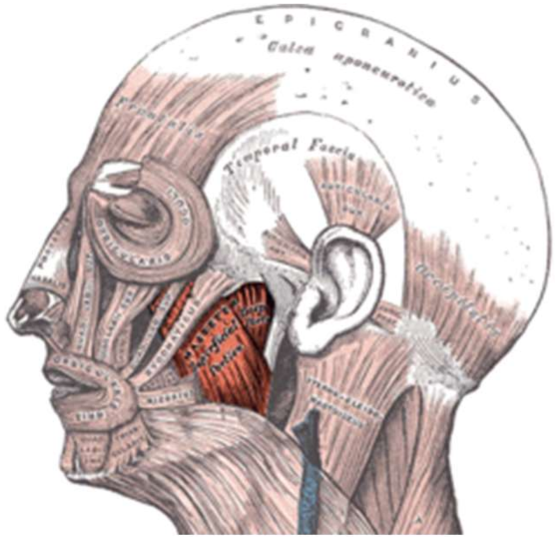

**Flexion**




Reference 1 modified

# MUSCLES OF MASTICATION

Myofascial Pain Referral




https://upload.wikimedia.org/wikipedia/commons/thumb/f/fa/Gray378_%28masseter_highlight%29.png/250px-Gray378_%28masseter_highlight%29.png

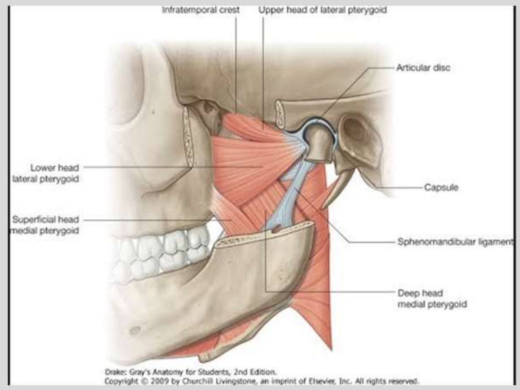
# Masseter

- Major function is closure of the jaw
- Hypertonicity/Trigger point symptoms:
  - Upper teeth pain
  - Lower teeth pain
  - Maxillary pain (which can be confused with sinusitis).
  - Ear pain
  - Reduced opening
  - Tinnitus

Relationship with temporal bone- tinnitus.

# Lateral Pterygoid

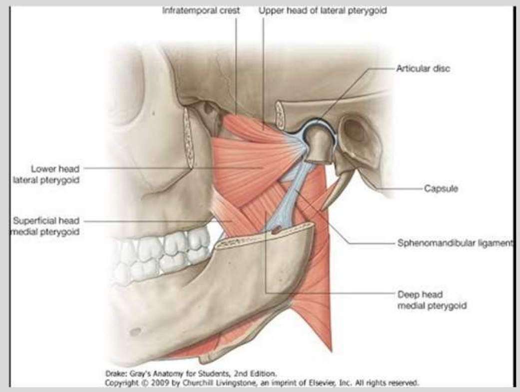
- Major functions is **OPENING THE JAW.**
- Unilateral contraction causes deviation to one side.
- **Inserts onto intra-articular disc**
- Hypertonicity/Trigger Point Signs/Symptoms:
  - Reduced opening
  - Altered occlusion
  - Maxillary region pain
  - TMJ pain/disc derangement
  - Clicking



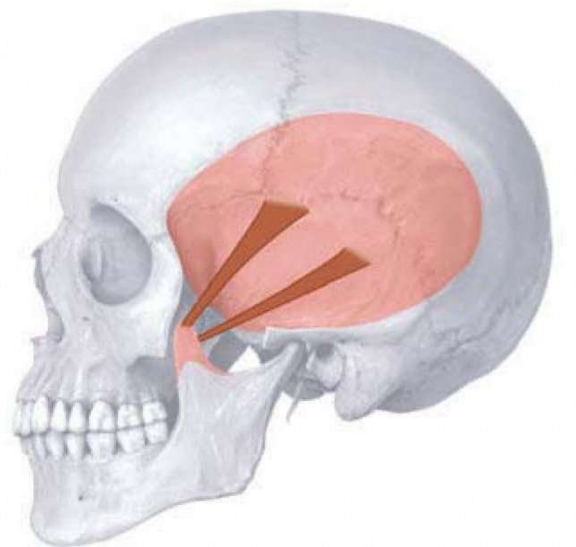

https://i.ytimg.com/vi/6-fOXJ1Nk2o/hqdefault.jpg

# Medial Pterygoid

- Major function: closure of jaw
- Unilateral contraction: deviation of jaw to one side
- Hypertonicity/Trigger points symptoms:
  - Mouth and throat pain
  - Painful swallowing
  - “Stuffiness” related to ear




https://i.ytimg.com/vi/6-fOXJ1Nk2o/hqdefault.jpg



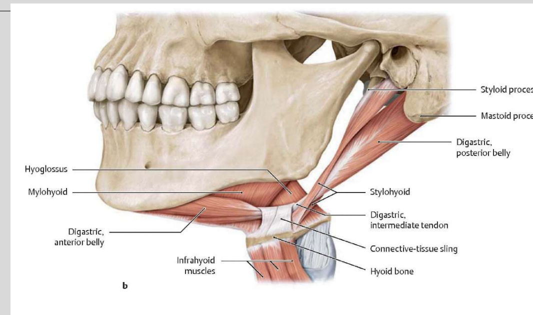
# Temporalis:

- Major function is closure of mandible.
- Hypertonicity/Trigger point signs/symptoms:
  - “Zig-zag” deviation pattern
  - Headache
  - Upper teeth pain

# Muscles of Oral Floor and Hyoid

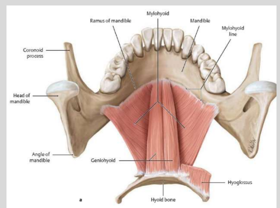
Suprahyoid Muscles

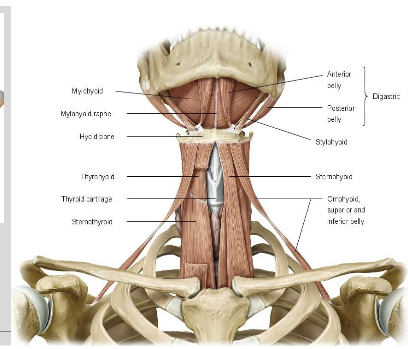
- Mylohyoid
- Digastric (Ant. & Post.)
- Geniohyoid
- Stylohyoid




Infrahyoid Muscles

- Sternohyoid
- Omohyoid (Sup. & Inf.)
- Thyrohyoid
- Sternothyroid




Reference 1

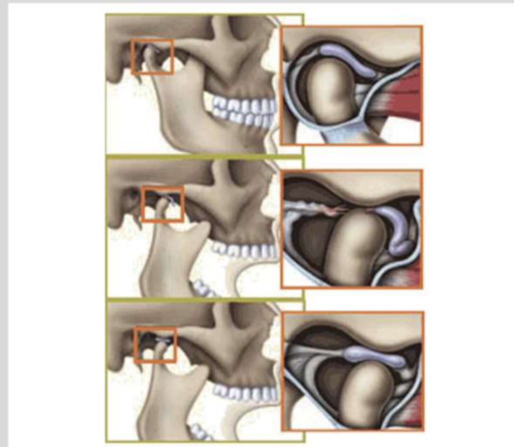
| Muscle | Origin | Insertion | Innervation | Function |
|---|---|---|---|---|
| Masseter | Zygomatic arch and maxillary process of the zygomatic bone | Lateral surface of ramus of mandible | Masseteric nerve from the anterior trunk of the mandibular nerve [$V_3$] | Elevation of mandible |
| Temporalis | Bone of temporal fossa and temporal fascia | Coronoid process of mandible and anterior margin of ramus of mandible almost to last molar tooth | Deep temporal nerves from the anterior trunk of the mandibular nerve [$V_3$] | Elevation and retraction of mandible |
| Medial pterygoid | Deep head—medial surface of lateral plate of pterygoid process and pyramidal process of palatine bone; superficial head—tuberosity of the maxilla and pyramidal process of palatine bone | Medial surface of mandible near angle | Nerve to medial pterygoid from the mandibular nerve [$V_3$] | Elevation and side-to-side movements of the mandible |
| Lateral pterygoid | Upper head—roof of infratemporal fossa; lower head—lateral surface of lateral plate of the pterygoid process | Capsule of temporomandibular joint in the region of attachment to the articular disc and to the pterygoid fovea on the neck of mandible | Nerve to lateral pterygoid directly from the anterior trunk of the mandibular nerve [$V_3$] or from the buccal branch | Protrusion and side-to-side movements of the mandible |

http://www.people.virginia.edu/~dgm4n/Lab03/Images/Table8.11_MasticationMuscles.jpg

# TMJD

# “Popping” and “clicking”

•Anterior displacement of disc relative to condyle.
•As condyle translates forward, it locates itself onto the central portion of the disc, and a **“click”** occurs.
•Translation then continues, allowing further jaw opening.
•As jaw closes, condyle moves off the disc and relocates behind the disc, causing a reciprocal **“click.”**

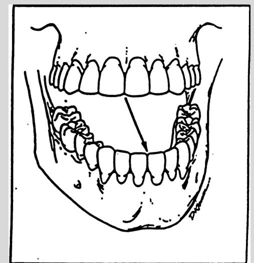


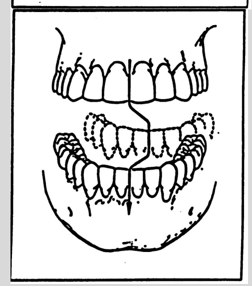
https://www.melbournetmjcentre.com.au/wp-content/uploads/2017/10/Clicky-or-Clicking-Jaw-TMJ-blog.jpg

# Abnormal Tracking

- Mandible not opening in a straight line
  - Deviation can occur to same or opposite side
- May first open to one side, then return to middle **(deflection)**

*Note: When unilateral medial pterygoid is involved, jaw can deviate to the opposite side*





# Jaw Locking

1. Clicking may progress to Locking
2. Condyle fails to locate itself on the disc
3. Disc blocks condylar movement
4. Results in **“closed lock”**
5. Condyle pushes disc forward as opening effort is made
6. Disc can deteriorate and change shape

- Locking can be acute or chronic
- Locking may reduce or progress to become persistent

# PHYSICAL EXAMINATION

# Evaluation

Observation

- Facial symmetry
- Mandibular deviation
- Measured jaw opening
- Joint click or pop

Palpation

- Joint palpation
- Muscle & ligament palpation
- PRM motion & quality, especially temporal bones

Further Assessment

- MRI if internal derangement suspected

# Treatment

- Acute:
  - Trauma (dental work)- consider OMT before and after dental work
  - Often self limiting
  - OMT
  - Patient education (self-massage, exercises, mindful of chewing, etc.)
- Chronic:
  - Team approach: dentist, Osteopathic physician, psychologist, etc.
  - OMT
  - Relaxation techniques
  - Muscle relaxants
  - Bite guards
  - Patient education and exercises.
  - Supplements/herbs
  - Avoid caffeine/alcohol
  - Acupuncture/biofeedback

# Osteopathic Treatment

- Mandible, Temporal Bone and Sphenoid Bone
- Consider muscles of mastication origin and insertion
- Muscles: Lateral Pterygoid Muscle Inhibition and Mandibular Muscle Energy
- Ligaments: Stylomandibular and Sphenomandibular ligament balancing
- Cervical somatic dysfunction
- Consider postural imbalance and leg length discrepancy
- Treat sacrum and pelvis somatic dysfunction

# Research

- Intraoral Manipulation and Jaw Exercises Shown to be Benefit in TMJ Disorder:
  - IMT (Intraoral Manipulation) and IMTESC (Intraoral Manipulation plus Exercise Protocol) groups both had statistically significant lower pain scores at 6 weeks, 6 months and 1 year assessments.
  - No difference between groups except at 1 year assessment with IMT has slightly higher pain scores.
- Manual Therapy for the Mandibular Accessory Ligaments for the Management of TMJ Disorders:
  - American Academy of Craniomandibular Disorders and Minnesota Dental Association have cited physical therapy (electrophysical modalities, therapeutic exercises, manual techniques) as an important treatment modality for the relief of musculoskeletal pain, reduction of inflammation, and restoration of oral motor functinon.
- Manual Therapy for Hamstring Hypertonicity Improves TMD in Athletes:
  - After hamstring stretching techniques, athletes had improved hamstring extensibility and active mouth opening.
- OMT effects on Mandibular Kinetics: Kinesiographic Study:
  - Study group (received OMT) showed significant improvement of maximal mouth opening and maximal mouth opening velocity compared with the no-intervention group.
- Effectiveness of OMT versus Osteopathy in the Cranial Field in TMJD-a pilot study

  Findings support the use of OMT and OCF and support an interdisciplinary collaboration in pt management.

# Objectives

1. Identify the emphasized cranial, muscular, ligamentous, & neural anatomy related to the pathophysiology of the following clinical cases:
   1. Headache
   2. TMJD
   3. **Bell's Palsy**
   4. **Vertigo/Dizziness**
2. Define the role of different treatments, including OCMM and other OMT techniques, as it relates to the pathology discussed and emphasized in the self-study
3. Compare and contrast normal motion and pathological motion of the individual cranial structures potentially associated with conditions discussed during the self-study.
4. Define relevant research as it relates to the conditions emphasized in the self-study.

# Case 3

- 72-year-old male presents reporting dizziness. He reports a sensation of whirling and unsteadiness, which occurs daily, lasting several seconds to a couple hours. It began 1 month ago shortly following a MVA in which he was stopped and was rear ended by a car going about 35 mph. He was wearing a seatbelt, experienced rapid flexion and extension of neck. No LOC and was not taken to the hospital. He started to experience neck pain and was diagnosed with whiplash. Neck movement can exacerbate symptoms. No visual or hearing changes, no numbness/tingling/weakness. No HA or nausea/vomiting. No congestion or recent infections.
- Meds: HCTZ and Ibuprofen for pain
- NKDA
- PMhx/PShx: Type II diabetes controlled with diet/exercise, HTN, Mild OA of knees. No surgeries
- Social Hx: Retired engineer, No smoking or illicit drug use, 4-5 beers a week.
- Family hx: non-contributory
- ROS: neg. except what is in HPI.

# Case 3

- VSS
- HEENT: EOMI, PERRLA, normal fundoscopic exam, nystagmus noted with body rotation
- CV/Lungs: Normal
- Neuro: CN II-XII grossly intake, DTR 2/4 b/l in UE and LE, MS 5/5 in UE and LE b/l, No atrophy, spasticity, clonus or tremors noted. Normal finger-to-nose, Rhomberg and heel-to-shin. Sensation and two point discrimination intact. Proprioception intact.
- OSE:
  - Head: OA FSlRr
  - Cervical: Tenderness/hypertonicity in suboccipital region and upper cervical spine, C3 FSRl
  - UE: right scapula elevated and tenderness over right levator scapulae insertion
  - Thoracic: Paraspinal muscle hypertonicity in upper thoracic region
  - Lumbar: Paraspinal muscle hypertonicity

# Differential Diagnosis of Dizziness

Not Comprehensive

- BPPV
- Stroke
- Migraine
- Medication Side Effect
- Dehydration
- Labyrinthitis
- Meniere's disease
- Acoustic Neuroma
- Cervicogenic vertigo
- Brain tumor

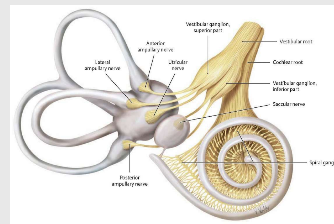
# Case 3

- Dizziness:
  - Nonspecific term used by patient to describe a variety of symptoms including lightheadedness, feeling faint, weak or unsteady.
- Vertigo:
  - Sensation of movement, usually spinning or rotating or whirling associated with vestibular system.
- Balance:
  - Whole body process
  - Three main components:
    - Inner Ear
    - Visual perception
    - Proprioceptive
  - Central process takes place in the vestibular nuclear complex and cerebellum

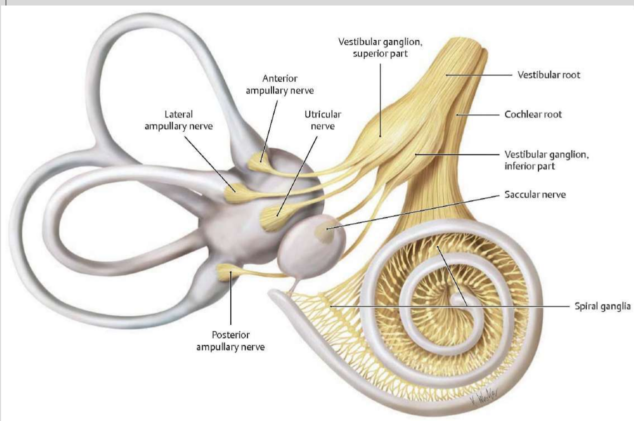
Is the room spinning??

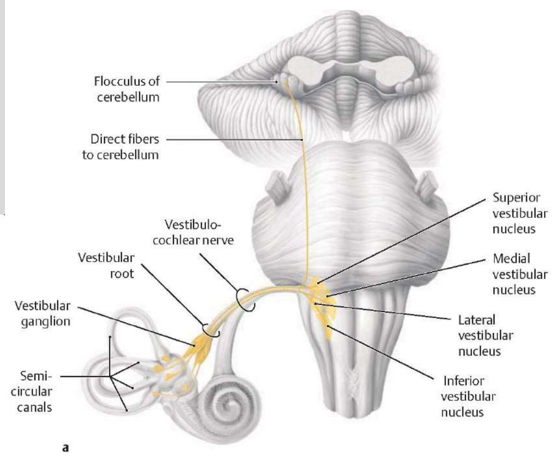
# CN VIII: Vestibulocochlear Nerve

- Origin:
  - Vestibular Nucleus (4)
  - Cochlear Nucleus (2)
- Fiber Types:
  - Special Somatic Afferent:
    - Vestibular: gravitational equilibrium
    - Cochlear: auditory
- Path:
  - Following their afferent pathway, *they share a common dural sheath* as they pass out the internal auditory meatus to their respective nuclei at the cerebellopontine angle
- Clinical/Osteopathic Considerations:
  - ***Synchronous physiologic temporal bone motion is critical for proper equilibrium information*** **as well as proper** ***pressure gradients of the inner, middle, and external ear. Dural strains can be present at the IAM to alter neural axonal health***
  - ***Tinnitus and vertigo are common symptoms***



# Vestibulocochlear Nerve



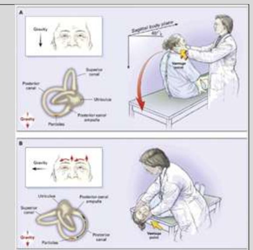




Illustrator: Karl Wesker pp. 82-83

Schuenke et al. THIEME Atlas of Anatomy • Head and Neuroanatomy

# Cervicogenic Vertigo

- Dizziness, unsteadiness and unbalance of whiplash injury
- Concurrent with neck pain and stiffness and worse with neck movement
- No hearing loss or tinnitus, possible nausea
- Probable etiology: Disturbance in cervical proprioceptive input to vestibular nuclear complex and cerebellum
- Several studies have shown manual treatment to be effective at improving symptoms of dizziness and multiple case reports have been published.
- Questionable diagnosis- should be treated similar to cervical pain/whiplash



# Common Diagnosis

- Acute Labyrinthitis
  - Alone or in combination with viral or bacterial infection
  - Hearing loss, tinnitus, nausea, vertigo and nystagmus
  - Acute episode last 24-48 hrs and slowly improves over several weeks
- Benign Paroxysmal Positional Vertigo (BPPV)
  - Debris in endolymph of posterior SCC
  - Presents as episodic vertigo with head position
  - Episode lasts 30 seconds to 2 minutes
  - <u>No hearing loss or tinnitus</u>
  - Dix-Hallpike positioning test
  - Self-limiting, vestibular exercises
- Meniere's Disease
  - Increased endolymphatic pressure
  - Hearing loss and tinnitus
  - Severe vertigo, nausea and vomiting (30 minutes to 24 hours)
  - Intensity diminishes over several days, unsteadiness may last weeks, tinnitus and hearing loss may be permanent
- Acoustic Neuroma
  - Schwannoma of eighth cranial nerve causes unilateral sensorineural hearing loss and tinnitus
  - Vertigo is not usually a symptom and is short lived if occurs

# Osteopathic Evaluation and Treatment

- Temporal Bone (common to have one external rotation and one internal rotation or asynchronous with respiration)
- SBS
- Occiput
- Tentorium Cerebelli
- SCM, clavicle and sternum
- Cervical Spine
- Sacrum/Pelvis
- Lower Extremity

# Research

- **Patients:** Sixteen participants (2 male, 14 female; mean [range] age, 49 [13-75] years) with dizziness lasting at least 3 months (mean duration of symptoms, 84 months) and spinal somatic dysfunction, but no history of known stroke or brain disease, were recruited from the local community and evaluated for postural balance control before, immediately after, and 1 week after OMT.
- **Intervention:** Four osteopathic physicians board certified in neuromusculoskeletal medicine/osteopathic manipulative medicine provided OMT, including muscle energy; high-velocity, low-amplitude; counterstrain; myofascial release; balanced ligamentous release; and cranial OMT techniques.
- **Main Outcome Measures:** Outcomes were assessed with the SMART Balance Master (NeuroCom), a validated instrument that provides graphic and quantitative analyses of sway and balance, and the Dizziness Handicap Inventory (DHI), a self- assessment inventory designed to assess precipitating physical factors associated with dizziness and functional and emotional consequences of vestibular disease.
- **Results:** Paired *t* tests, performed to assess changes in mean composite scores for all challenge tests, revealed that balance was significantly improved both immediately and 1 week after OMT (both $P<.001$), with no significant difference between immediate and 1-week post-OMT scores ($P=.20$). The DHI scores, both total and subscale, improved significantly after OMT ($P<.001$), and changes in composite and DHI scores were correlated with each other ($P=.047$).
- **Conclusion: Osteopathic manipulative treatment for spinal somatic dysfunction improved balance in patients with dizziness lasting at least 3 months.**


- Marcel Fraix, M. Gordon, A. Graham, V. et. al. *Use of the SMART Balance Master to Quantify the Effects of Osteopathic Manipulative Treatment in Patients With Dizziness. J Am Osteopath Assoc.* 2013;113(5):394-403

# Case 4

- The patient in the present case was a **26-year-old woman** who arrived at the osteopathic manipulative medicine clinic at the University of North Texas Health Science Center at Fort Worth—Texas College of Osteopathic Medicine with **right-sided facial weakness**, which she said she had for **1 week**, accompanied by a **dulled sense of taste**. She noted that the day before the onset of her symptoms, she was up all night studying.
- In response to questions, she said she had not experienced trauma or fallen asleep in a drafty room. Nor, she claimed, did she have previous infection with herpes simplex virus–type 1, acute or chronic auditory disorders, or recent dental work. The patient's medical history was notable for an incident of **facial weakness that lasted for 1 month when she was 7 years old**. The only surgical procedure in her medical history was a tonsillectomy. During the week prior to her visit to the clinic, she had noticed only slight improvement of her symptoms. She said she was not taking any medication.
- ***Physical Examination***
  - **The patient's physical examination revealed weakness of her right eyelid with delayed blinking, along with a mild right-sided facial droop and lips that remained flat on the right side when she attempted to smile**. Cranial nerves II through XII were intact, except for the right facial nerve (CN VII). The patient's upper and lower extremity deep-tendon reflexes, muscle strength, and sensation were grossly intact.
- ***Structural Examination***
  - The patient's general appearance was most notable for facial asymmetry with decreased tone on the right side. The structural examination also revealed a **side-bending/rotation dysfunction** on the right side of the cranium. Cervical vertebrae **C2 and C5 were side-bent and rotated right**. **Fascial restrictions were found in the patient's cervical and thoracic outlet areas**. Standing and seated flexion tests revealed iliosacral and sacroiliac somatic dysfunction on the right side. There was **decreased right sacroiliac motion and right anterior innominate rotation.**

Lancaster, D. Crow, W. *Osteopathic Manipulative Treatment of a 26-Year-Old Woman With Bell's Palsy. JAOA.* Vol106. No5. pp. 285-289.

# Case 4

- Osteopathic manipulative treatment (applied in two sessions, each lasting approximately 20 minutes)
- Treatment focused on enhancing the patient's lymphatic circulation.
- No medications were prescribed.
- The four key diaphragms were treated to aid lymphatic circulation: the thoracic outlet, the respiratory diaphragm, the suboccipital diaphragm, and the cerebellar tentorium.
- Somatic Dysfunctions of the pelvis and sacrum were treated.
- Thoracic Pump
- Cervical Spine SD were treated
- Mandibular Drainage Technique (Gallbreath Treatment)
- OCMM to balance Temporal Bones (Tentorium Cerebelli)

Lancaster, D. Crow, W. *Osteopathic Manipulative Treatment of a 26-Year-Old Woman With Bell's Palsy. JAOA.* Vol106. No5. pp. 285-289.

# Case 4

- The morning after her treatment, the patient reported that most of her muscle tone had returned and her sense of taste was again normal.
- On the third day posttreatment, there was a 24-hour exacerbation of the symptoms after the patient consumed two alcoholic drinks.
- Within 1 week after treatment, however, she had fully regained her sense of taste and almost all of her facial tone. Only a mild delay in blinking remained.
- The patient's second treatment session, which took place 1 week after the first session, included all of the OMM procedures described above, except for muscle energy technique for the cervical area.
- One week after the second treatment session, the patient was asymptomatic.

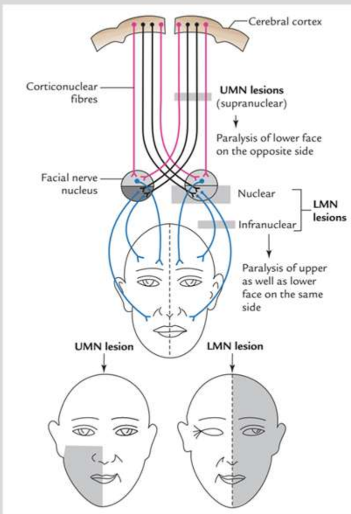
Lancaster, D. Crow, W. *Osteopathic Manipulative Treatment of a 26-Year-Old Woman With Bell's Palsy. JAOA.* Vol106. No5. pp. 285-289.

# Bell's Palsy

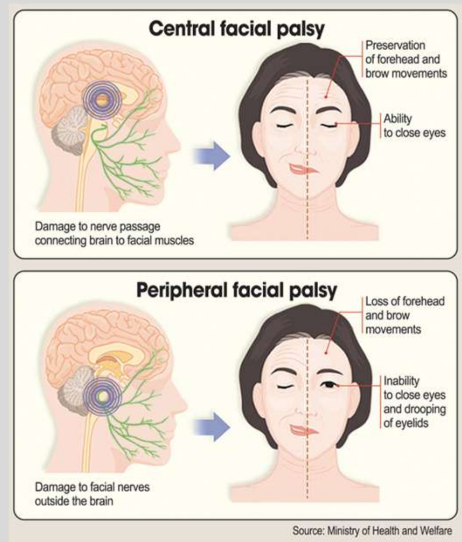
- Unilateral facial nerve paralysis (Bell's Palsy) is an idiopathic, acute condition caused by a lesion of the facial nerve, resulting in unilateral paralysis or paresis of the face.
- Approximately 23 in 100,000 persons per year become afflicted by Bell's palsy.
- Symptoms, which can affect either side of the face, typically first occur in patients between the ages of 10 and 40 years.
- Bell's palsy affects men and women at a roughly equal rate of occurrence.
- Physiology involves inflammation of the facial nerve within the osseous facial canal, causing compression and ischemia of the nerve.
- A viral etiology is suspected
- Symptoms: facial muscle weakness/paralysis (including forehead), change in taste, hyperacusis and pain behind the ear, unable to open eyelid and decreased tear production.

Lancaster, D. Crow, W. *Osteopathic Manipulative Treatment of a 26-Year-Old Woman With Bell's Palsy. JAOA.* Vol106. No5. pp. 285-289.

# Facial Nerve

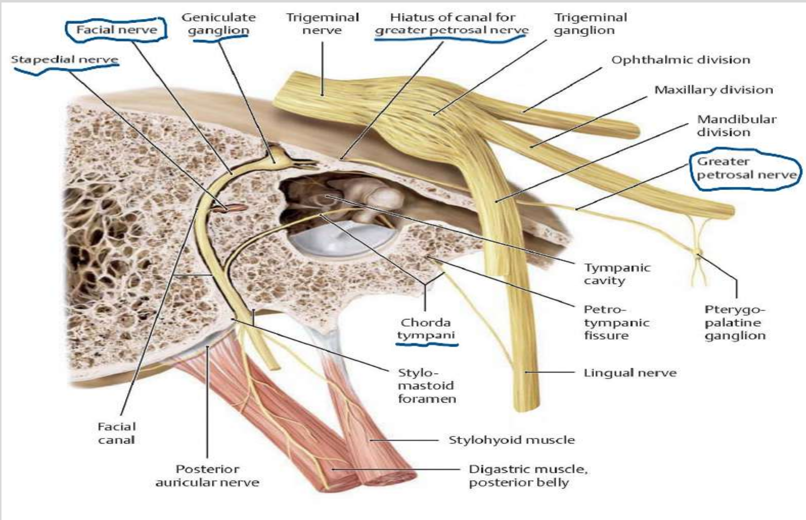



https://www.google.com/url?sa=i&source=images&cd=&ved=2ahUKEwjh4p_vxcvjAhUIKKwKHRE_BAoQjRx6BAgBEAU&url=https%3A%2F%2Fwww.earthslab.com%2Fanatomy%2Ffacial-nerve%2F&psig=AOvVaw14BNPf9rlbSEfppEy_hnHd&ust=156398863407711
1




https://www.google.com/url?sa=i&source=images&cd=&ved=2ahUKEwjQhsi7xcvjAhUBQq0KHT-WBn0QjRx6BAgBEAU&url=https%3A%2F%2Fwww.koreatimes.co.kr%2Fwww%2Fculture%2F2019%2F03%2F641_200432.html&psig=AOvVaw1XpdhDi0-sQBDNTLiYmGId&ust=1563988525410878

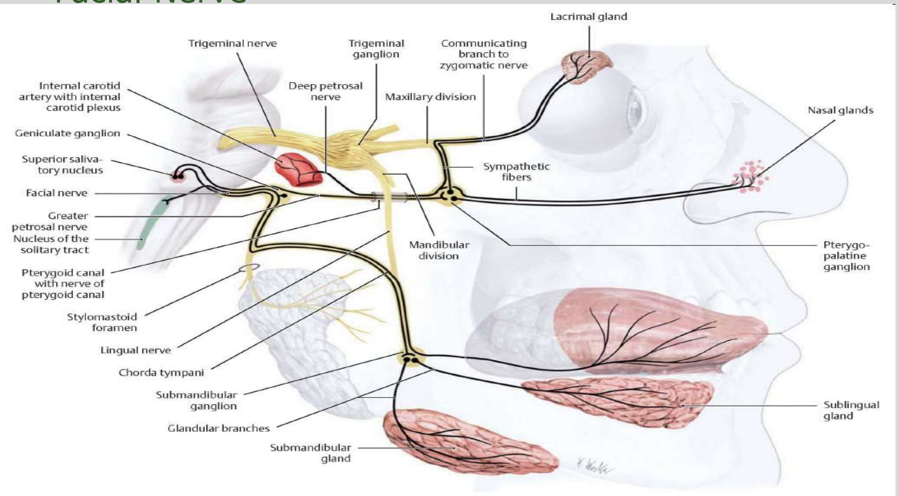
## Facial Nerve Considerations within the Petrous Portion of the Temporal Bone




Illustrator: Karl Wesker pp. 78-79
Schuenke et al. THIEME Atlas of Anatomy • Head and Neuroanatomy

Thieme

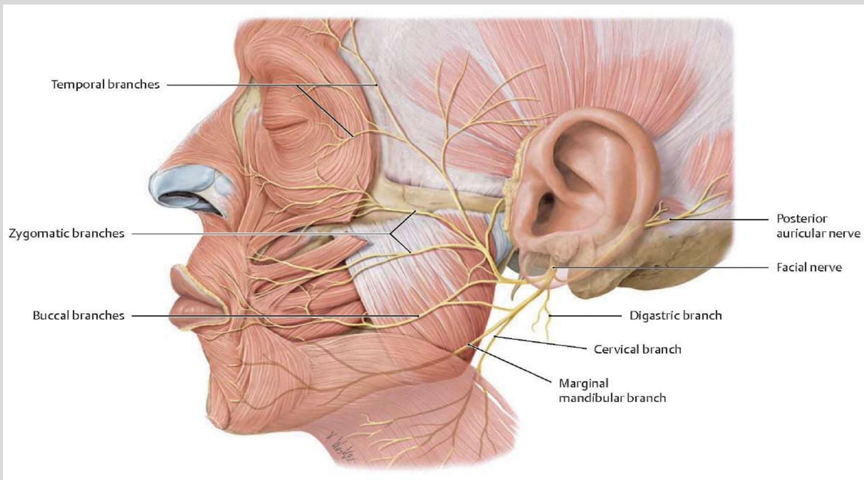
## Parasympathetic & Sympathetic Efferents of the Facial Nerve




**C Parasympathetic visceral efferents and visceral afferents (gustatory fibers) of the facial nerve**

Illustrator: Karl Wesker pp. 78-79

# Facial Muscle Innervations




Illustrator: Karl Wesker pp. 78-79

Schuenke et al. THIEME Atlas of Anatomy • Head and Neuroanatomy



Thieme

# Osteopathic Examination and Treatment

- Temporal Bone
- Cervical Spine
- Myofascial restriction in head and neck
- Diaphragms/lymphatic treatments
- Pelvis/Sacrum
- Upper Thoracic Spine

# References


- Chila, A. Foundations of Osteopathic Medicine. 3rd Edition. Lippincott Williams and Wilkins. 2011: pp. 485-512, 910-917.
- Digiovanna, E. Schiowitz, S. Downling, D. An Osteopathic Approach to Diagnosis and Treatment. 3rd Edition. Lippincott Williams and Wilkins. 2005: pp. 577-582, 578.
- Nelson, K. Glonek, T. Somatic Dysfunction in Osteopathic Family Medicine. 2nd ed. 2015: 201-215, 234-242.
- Espejo-Antunex L, et.al. *Manual Therapy for Hamstring Hypertonicity Improves Temporomandibular Dysfunctions in Athletes.* J. Bodyw Mov Ther. Doi:10.10.16/j.jbmt.2015.12.012.
- Kalamir A, et.al. *Intraoral Manipulation and Jaw Exercises Show to be of Benefit in Temporomandibular Disorders.* J. Manipulative Physiol Ther. 2012;35(1): 26-27. JAOA.
- Monaco A. et.al. *Osteopathic Manipulative Treatment Effects on Mandibular Kinetics: Kinesiographic Study.* European Journal of Pediatric Dentistry; 1/2008. p. 37-42.
- Cuccia A. et.al. *Manual Therapy of the Mandibular Accessory Ligaments for the Management of Temporomandibular Joint Disorders.* JAOA. 2011;111(2):102-112.
- Melissa Pearce, DO Powerpoint: Presentation: *Osteopathic Considerations of the Temporomandibular Joint (TMJ)*
- Mitch Hiserote, DO Powerpoint Presentation: *Osteopathic Considerations of the Temporal Bone*
- Stacey Pierce-Talsma, DO Powerpoint Presentation: *OCMM3: Introduction to Diagnosis: Dysfunction of the SBS*
- Stacey Pierce-Talsma, DO Powerpoint Presentation: *OCMM6: The Sphenoid*
- Jordan Keys, DO Powerpoint Presentation: *OCMM5: Clinical Considerations I*
- Vanessa Newman, DO Powerpoint Presentation: *OCMM5: Clinical Considerations I*

Gesslbauer C, Vavti N, Keilani M, Mickel M, Crevenna R. Effectiveness of osteopathic manipulative treatment versus osteopathy in the cranial field in temporomandibular disorders - a pilot study. Disabil Rehabil. 2018 Mar;40(6):631-636. doi: 10.1080/09638288.2016.1269368. Epub 2016 Dec 28. PMID: 28029069.